

Objectives: To investigate the alignment between self-management needs and strategies used by individuals with traumatic brain injury (TBI) and training provided by the Chronic Disease Self-Management Training Program (CDSMP) to determine if the training aligns with the needs of people with TBI.

Design: The study involved a secondary analysis of qualitative data from interviews conducted with individuals with TBI. The interviews focused on their experiences with community participation after TBI, including areas of work, school, community life, recreation & leisure, and relationships.

Setting: Interviews occurred via a secure video conference platform.

Participants: Interview participants included 22 community-dwelling individuals with a history of TBI. The average age was 48.6 years (SD 14.1), the average time since injury was 4.4 years (SD 2.8), and 68% had “mild” TBI.

Interventions: Not applicable.

Main Outcome Measures: A codebook was developed based on the core self-management skills that form the foundation of the CDSMP. These skills include problem solving, decision making, using resources, relationships with health care providers, and taking action. Additional codes were added inductively based on the interview data.

Results: Several interview participants reported using self-management strategies to help support their participation, whereas others reported barriers to participation that aligned with self-management training, indicating a potential training need. In addition to the self-management skills that underlie the CDSMP, interpersonal relationship skills were identified as both important and a barrier to community participation.

Conclusions: The CDSMP self-management training skills appear to align with self-management strategies and needs identified by individuals with TBI. The results of this study suggest that future research should examine the effectiveness of CDSMP for individuals with TBI. Additional training or TBI-specific adaptations may be beneficial to address TBI-specific needs; for example, managing interpersonal relationships outside of a health care setting.

Disclosures: none.

Keywords: Brain injury, Brain damage, Chronic, Chronic disease, Rehabilitation, Self-management

Research Poster 2776563

Capturing Deficits at Diagnosis: The Case for Early Speech-language Pathology Assessment of Pediatric Patients with Nonneural Cancer

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Objectives: To ascertain the prevalence of deficits within the Speech-Language Pathology (SLP) scope of practice in children recently diagnosed with nonneural cancers.

Design: Retrospective review of speech, language, feeding, swallowing, and cognitive function surveyed via a battery of paper-based screening instruments administered at cancer diagnosis. Demographic information was extracted from the electronic health record. Descriptive statistics were used to summarize participant characteristics and screening survey results.

Setting: Inpatient oncology units at free-standing, academic-affiliated health care system in the Southern United States.

Participants: Fifty patients between 9 months and 19 years assessed within 7 days of nonneural cancer diagnosis.

Interventions: Not applicable.

Main Outcome Measures: The battery of screening instruments included: Patient-Reported Outcomes Measurement Information System (PROMIS) Pediatric Cognitive Function-Short Form 7a, Quality of Life in Neurological Disorders (Neuro-QOL) Pediatric Cognitive Function-Short Form, National Institute on Deafness and Other Communication Disorders Speech and Language Developmental Milestones, Neonatal Eating Assessment Tool Screening Instruments (NeoEAT), Pediatric Eating Assessment Tool Screening Instruments (PediEAT), and the Children's Health Speech Therapy Oncology Screening questionnaire.

Results: At cancer diagnosis, 20% of children had prior SLP intervention, and 26% had a history of disorders within the SLP scope of practice. Fifty percent of caregivers reported their children had not attained milestones, particularly in word acquisition, word chaining, and speech sounds. Results of the Feeding Flock Screening instruments indicate that 65% of children had behaviors necessitating SLP assessment consistent with sensory aversions, oral motor delays, and feeding refusal. PROMIS and Neuro-QOL screener results concurrently indicated cognitive deficits in 30% of children.

Conclusions: Caregivers reported over 1 in 4 children had disorders within the SLP scope of practice at cancer diagnosis, higher than the 6-15% documented in the noncancer pediatric population. Our findings suggest the prevalence of disorder may be even higher, with 50% of children failing to meet speech-language milestones, 65% with concerning feeding behaviors, and 30% with cognitive deficits, possibly reflecting more recent functional change between cancer onset and diagnosis. While deficits in children with nonneural cancers are commonly attributed to the sequelae of neurotoxic treatment and abnormal developmental environments, their presence at cancer diagnosis suggests there may be other influencing factors. Further investigation is needed to determine the presence of deficits in patients not hospitalized at diagnosis as well as the benefits of early screening, SLP treatment, and other supportive care interventions in mitigating the impact of developmental, feeding, and cognitive deficits experienced by patients with childhood cancer.

Disclosures: none.

Keywords: Inpatients, Neoplasms, Patient-reported outcome measures, Prevalence, Rehabilitation, Speech-language pathology

Research Poster 2751202

Lower Extremity Youth Volleyball Injuries Predominantly Impact Female Athletes: A Sex-specific Analysis of National Injury Data from 2013 to 2022

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Objectives: To evaluate current youth volleyball player lower extremity (LE) injury trends and sex-specific differences in the United States.

Design: Statistical analysis of the National Electronic Injury Surveillance System database.

Setting: All emergency departments across the United States.

Participants: Patients ≤ 18 years old who visited the emergency department for a LE volleyball injury from January 1, 2013 to December 31, 2022.

Interventions: Not applicable.

Main Outcome Measures: Body part injured, diagnosis, mechanism of injury, age, gender, annual trends in injury, and disposition (national estimates were calculated using the hospitals' corresponding statistical sample weights).

Results: A total of 134,785 (raw cases, 3957) LE recreational and/or competitive indoor volleyball-related injuries were included. Average age was 14.8 ± 2.0 years. Females made up 80.6% of patients. The most common body parts injured were ankles (78,315, 58.1%), knees (36,202, 26.9%), and feet (11,419, 8.5%). The most common identifiable mechanisms of injury were twisting (73,314, 54.4%), impact with the floor (17,283, 12.8%), and impact with another player (9916, 7.4%). Most common diagnoses were strain/sprain (87,473, 64.9%), other/not stated (21,950, 16.3%), and fractures (10,675, 7.9%). During the study period, excluding 2020, there was a significant annual increase in overall injuries (129; $P = .004$; CI, 42-216). A greater proportion of females sustained strains/sprains (66.8% vs 61.6%; $P < .001$) and contusions/abrasions (5.9% vs 5.6%; $P = .046$), while a greater proportion of males sustained fractures (12.5% vs 7.0%; $P < .001$) and dislocations (5.5% vs 3.5%; $P < .001$).

Conclusions: LE injuries in volleyball have been increasing since 2013, mostly affecting the ankles and female youth players. Strains/sprains, pain and swelling, and fractures are the most common diagnoses, with females

experiencing more strains/sprains and contusions/abrasions and males experiencing more fractures and dislocations. The increase in LE volleyball injuries may be due to increased popularity of the sport and more athletes playing without ankle braces and/or less participation in plyometric training/ankle-strength conditioning involving jumping and landing safely.

Disclosures: none.

Keywords: Lower extremity, Mechanism of action, NEISS, Pediatric, Rehabilitation, Volleyball

Research Poster 2751185

Psychiatric Comorbidities Complicate a Neurologic Diagnosis: A Case Report Highlighting Interdisciplinary Collaboration

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Erika Trovato, Julieanne Shulman, Rachel Feld-Glazman**

Objectives: To illustrate the value of interprofessional collaboration and the use of objective measures for a patient undergoing inpatient rehabilitation with serious psychiatric comorbidities.

Design: Case Study.

Setting: Inpatient Rehabilitation Hospital.

Participants: A 22-year-old female with a history of obsessive-compulsive disorder, severe anxiety, and major depressive disorder with psychotic features was admitted from an inpatient psychiatric hospital to an acute inpatient rehabilitation facility (IRF) with an admitting diagnosis of Functional Neurologic Disorder (FND).

Interventions: The patient participated in standard treatment for FND, including physical therapy, occupational therapy, music therapy, neuropsychology, and psychiatry. Despite the interdisciplinary approach, functional mobility did not progress as expected. On exam, the patient presented with absent reflexes, decreased range of motion, and no response to electrical stimulation. The patient's atypical presentation and interdisciplinary collaboration prompted an electromyography (EMG) study. Results were consistent with diffuse polyaxonal neuropathy. She was subsequently transferred to an acute care hospital for further evaluation.

Main Outcome Measures: Functional performance, deep tendon reflexes, range of motion, EMG results.

Results: The patient presented to an IRF with the diagnosis of FND, with functional goals of performing functional mobility without skilled assistance. Physical exam findings, lack of response to electrical stimulation, and EMG findings prompted further workup, which revealed the presumed diagnosis of Guillain-Barré Syndrome. The patient received intravenous immunoglobulin with good effect. Upon return to IRF, updated care plans were implemented to focus on facilitating a safe discharge home, using a wheelchair, with less focus on psychiatrically based interventions.

Conclusions: This case study exemplifies the value of interdisciplinary collaboration to provide appropriate and effective patient care. It also highlights the complexities of working with patients who have rarer diagnoses and psychiatric comorbidities. It demonstrates the importance of using objective testing, even after the initial neurological workup in the acute setting. Finally, this case raises attention to the issue of implicit bias in health care generally and provides an opportunity for acute rehabilitation teams to consider protocols to combat it.

Disclosures: none.

Keywords: GBS, Inpatient rehabilitation, Interprofessional collaboration, Missed diagnosis, Psychiatric comorbidities

Research Poster 2724464

Sociodemographic and Injury Characteristics Contribute to Ethnoracial Differences in Societal Participation After TBI

**Rylea Ranum (University of Houston), Michael Williams,
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Objectives: To investigate contextual factors that may contribute to ethnoracial differences in societal participation following traumatic brain injury (TBI), which is a common neurological condition and has varied symptom presentations. Participation captures reintegration into one's community and meaningful social roles. Prior literature indicated worse functional outcomes among Black and Hispanic individuals relative to White individuals.

Design: This project was a secondary analysis of a de-identified archival data set (Sherer et al., 2015).

Setting: The sample consisted of people with TBI recruited from the general community.

Participants: The sample included 222 non-Hispanic White, 188 non-Hispanic Black, and 76 Hispanic participants with chronic TBI. The original study recruited its sample from TBI Model Systems study sites in Texas, Michigan, and Alabama.

Interventions: Not applicable.

Main Outcome Measures: The primary outcome was societal participation as measured by the Participation Assessment with Recombined Tools-Objective (PART-O) Total score.

Results: There were significant group differences across sociodemographic and clinical characteristics (age, education years, marital status, preinjury employment, preinjury income, and time since injury), which were used as covariates in the evaluation of ethnoracial differences in PART-O Total score. There was a consistent, larger proportion of males relative to females across ethnoracial groups. There was a statistically significant difference in unadjusted PART-O Total between racial and ethnic groups as determined by one-way analysis of variance ($F(2,483)=13.780$, $P<.0001$). When considering covariates, race was not a significant contributor to PART-O Total score ($P=.357$). Age, education, and marital status were significant contributors ($ps<.001$) in the adjusted model. Preinjury employment and income, along with time since injury, were not significant in the model.

Conclusions: Apparent ethnoracial differences in societal participation among people with TBI are at least partially driven by sociodemographic and injury characteristics. More specifically, baseline characteristics of age, education, and marital status are meaningful contributors to differences in participation, greater than race/ethnicity. Societal participation, and its potential drivers, require further investigation among ethnoracial minoritized groups to promote health equity in TBI outcomes.

Disclosures: Dr Williams discloses grant funding from the National Institute on Disability, Independent Living, and Rehabilitation Research.

Keywords: Community integration, Racial groups, Rehabilitation, Traumatic brain injury

Research Poster 2751186

Dimensionality of Well-being in People with Multiple Sclerosis: A Hierarchical Confirmatory Factor Analysis

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Objectives: To measure well-being among people with MS with psychometrically sound measurements so that the clinical impact of MS can be better understood. This study aimed to examine the psychometric properties and factor structure of the PERMA-Profil as a measure of well-being in people with MS.

Design: Cross-sectional study design.

Setting: Participants were recruited from the National MS Society.

Participants: N=373 participants. Most participants were females (81.0%) and Caucasians (90.6%). The inclusion criteria for this study included being at least 18 years old and having an MS diagnosis.

Interventions: Not applicable.

Main Outcome Measures: Well-being was assessed with the PERMA-Profil (Butler & Kern, 2016). Disability-related stress was measured using the Brief Disability-Related Stress Scale (BDRSS; Lee & Chan, 2022). Functional limitations were assessed using the World Health Organization Disability Assessment Schedule 2.0 (Ustun et al., 2010).